

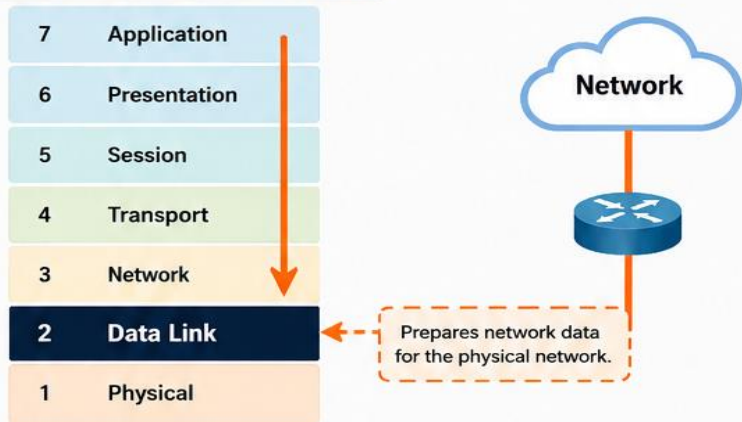
PURPOSE OF DATA LINK LAYER (OSI LAYER 2)

The Data Link Layer prepares network data for the physical network and controls reliable delivery between two directly connected nodes.

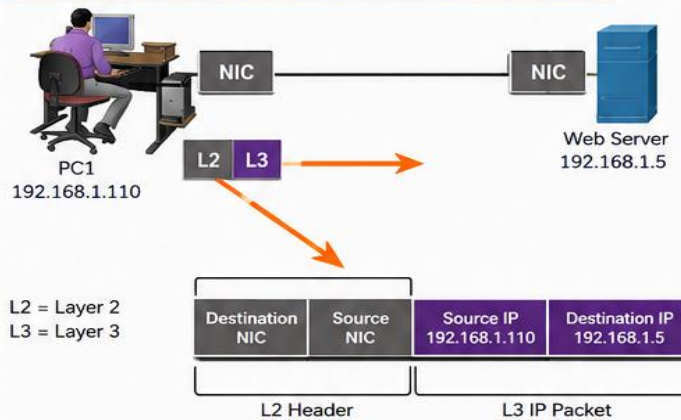
01 ROLE OF DATA LINK LAYER

-  Enables upper layers to access the media. Upper layers are unaware of the type of media used.
-  Accepts data (usually Layer 3 packets – IPv4/IPv6) and encapsulates them into Layer 2 frames.
-  Controls how data is placed on the media and received from the media.
-  Exchanges frames between endpoints over the network media.
-  Receives encapsulated data (usually Layer 3 packets) and delivers it to the correct upper-layer protocol.
-  Performs error detection and rejects any corrupt frame.

02 POSITION IN OSI MODEL



03 EXAMPLE – HOW DATA LINK LAYER WORKS



- 1 Data from the upper layer (L3 packet) is received by Layer 2.
- 2 Layer 2 adds its header (Destination NIC, Source NIC) to create a frame.
- 3 The frame is sent to Layer 1 (Physical layer) to be transmitted.

04 IEEE 802 LAN/MAN DATA LINK SUBLAYERS

IEEE 802 defines two sublayers in the Data Link Layer.

Network		Network Layer Protocol		
Data Link	LLC Sublayer	LLC Sublayer – IEEE 802.2		
	MAC Sublayer	Ethernet IEEE 802.3	WLAN IEEE 802.11	WPAN IEEE 802.15
		Standards for Fast Ethernet, Gigabit Ethernet, etc.	Standards for different types of wireless communications	Standards for Bluetooth, RFID, etc.
Physical				

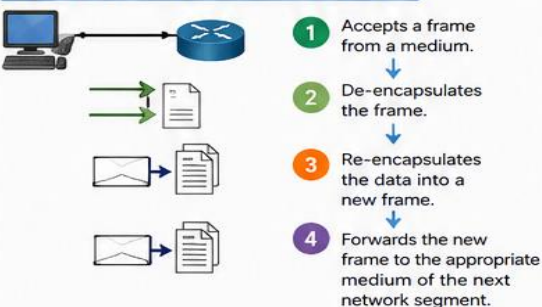
LLC Sublayer (IEEE 802.2)

- Takes network data (IPv4/IPv6 packet).
- Adds Layer 2 control information.
- Delivers the packet to the destination node.
- Controls the NIC and other hardware.

MAC Sublayer (Media Access Control)

- Provides data encapsulation.
 - Frame delimiting (marks start and end of frame)
 - Addressing (source & destination addresses)
 - Error detection (detects transmission errors)
- Provides media access control.
 - Allows multiple devices to share the medium (half-duplex).
 - Full-duplex communication does not require access control.

05 PROVIDING ACCESS TO MEDIA



KEY POINT
At each hop (router), Layer 2 performs these functions to move data from one link to the next.

06 DATA LINK LAYER STANDARDS

Various organizations define open standards and protocols for the Data Link Layer.

- IEEE – Institute of Electrical and Electronics Engineers
- ITU – International Telecommunication Union
- ISO – International Organization for Standardization
- ANSI – American National Standards Institute

ENGINEERING ORGANIZATION LOGOS



TOPOLOGIES (PART 1) – PHYSICAL AND LOGICAL

1. OVERVIEW

The Data Link Layer prepares network data for the physical network. It must know the logical topology of a network to transfer frames from one device to another.

Topology explains how devices (and intermediate devices) are arranged and interconnected.

There are two types of topologies used in LAN and WAN networks:

KEY POINT



- Physical topology = actual physical connections.
- Logical topology = how data flows (virtual connections).

2. TYPES OF TOPOLOGIES



Physical Topology

- Identifies the physical connections and how end devices and intermediate devices (routers, switches, wireless access points) are interconnected.
- Identifies the physical location of equipment (e.g., racks, rooms, shelves).



Logical Topology

- Refers to the way network transfers frames from one node to the next.
- Identifies virtual connections using device interfaces and Layer 3 addressing schemes.

3. DATA LINK LAYER AND TOPOLOGY



The data link layer “sees” the logical topology when controlling data access to the media.



It is the logical topology that influences the type of network framing and media access control used.

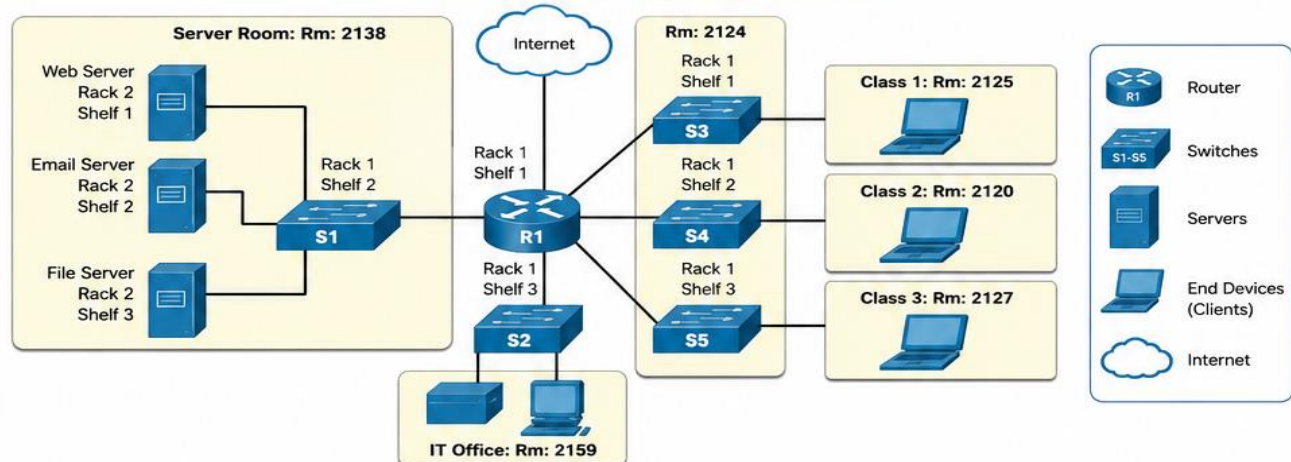


Different logical topologies can exist over the same physical topology.



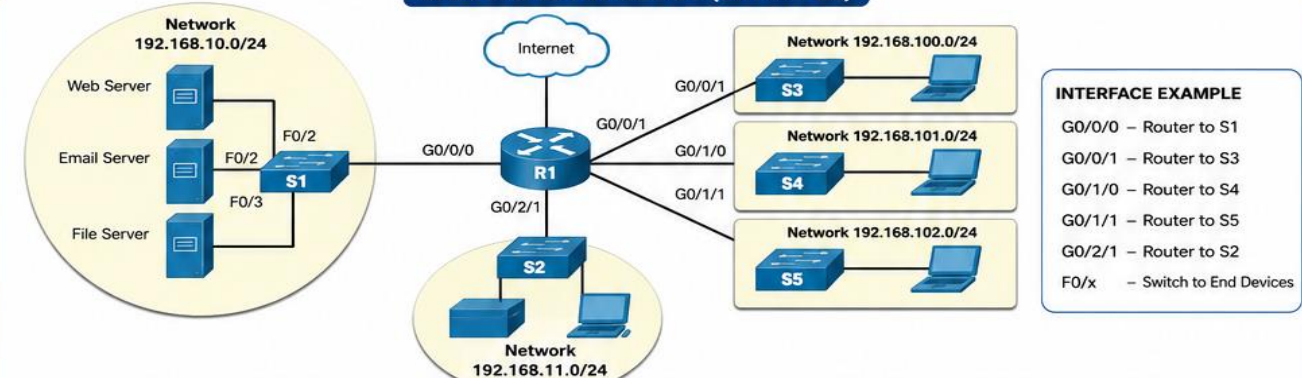
Understanding both topologies helps in efficient data transfer and troubleshooting.

4. PHYSICAL TOPOLOGY (EXAMPLE)



★ **Note:** This diagram shows the actual physical connections and locations of devices.

5. LOGICAL TOPOLOGY (EXAMPLE)



INTERFACE EXAMPLE

- G0/0/0 – Router to S1
- G0/0/1 – Router to S3
- G0/1/0 – Router to S4
- G0/1/1 – Router to S5
- G0/2/1 – Router to S2
- F0/x – Switch to End Devices

★ **Note:** This diagram shows how data flows logically using interfaces and IP addressing.

KEY TAKEAWAY

- Physical topology = where devices are and how they are connected.
- Logical topology = how data moves across the network.

TOPOLOGIES (PART 2)

WAN TOPOLOGIES & LAN TOPOLOGIES

1. WAN TOPOLOGIES

Wide Area Networks (WANs) connect sites over large geographical areas using different physical topologies.

1.1 COMMON WAN PHYSICAL TOPOLOGIES

Point-to-Point

Hub and Spoke

Mesh

A. POINT-TO-POINT (Simplest WAN topology)

- Consists of a permanent link between two endpoints.
- Simple and most common WAN topology.



B. HYBRID TOPOLOGY

- Combination of two or more topologies.
- Example: Partial mesh – some, but not all, devices are interconnected.

1.2 POINT-TO-POINT WAN TOPOLOGY (DETAILS)

- Directly connects two nodes (no other nodes in between).
- Common WAN protocol: Point-to-Point Protocol (PPP).
- Nodes can only communicate with the other endpoint.
- Places frames on the media at one end and receives them at the other.



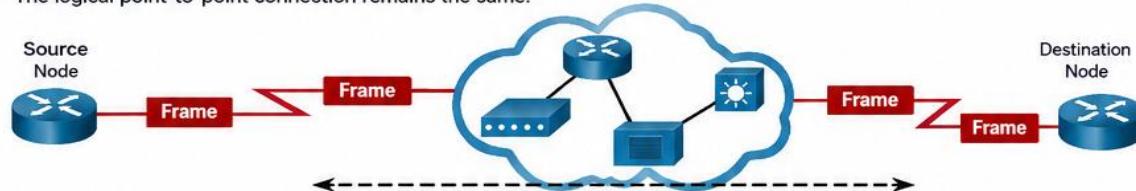
NOTE



- Device must know if an incoming frame is for it.
- Source and destination may be connected indirectly using intermediate devices.
- Using physical devices does not change the logical point-to-point connection.

1.3 POINT-TO-POINT CONNECTION WITH INTERMEDIATE DEVICES

- Intermediate devices (e.g., routers, switches) forward frames between source and destination.
- The logical point-to-point connection remains the same.



2. LAN TOPOLOGIES

2.1 STAR TOPOLOGIES

- Devices (nodes) connect to a central device (hub or switch) using individual links.
- Easy to install, manage, and troubleshoot.
- Failure of one link affects only that device.
- Most common LAN topology today.

2.2 EXTENDED STAR TOPOLOGY

- Multiple star topologies connected together.
- Uses multiple switches.
- Scalable and suitable for larger networks.

KEY TAKEAWAYS



- Physical topology = actual physical connections of devices.
- Logical topology = how data flows between devices.
- WANs use Point-to-Point, Hub-and-Spoke, Mesh (or hybrid) topologies.
- LANs use Star (most common today), Extended Star, Bus (legacy), and Ring (legacy) topologies.

2.3 LEGACY LAN TOPOLOGIES

A. BUS TOPOLOGY

- All devices share a single backbone cable.
- Terminators are used at both ends.
- Easy and inexpensive but failure of the backbone affects the whole network.

B. RING TOPOLOGY

- Each device connects to two neighbors forming a ring.
- Data travels in one direction (or both).
- A break in the ring can stop communication (unless dual ring).

2.4 COMMON PHYSICAL LAN TOPOLOGIES (DIAGRAMS)

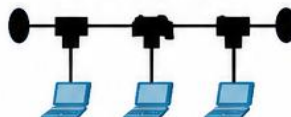
A. STAR TOPOLOGY



B. EXTENDED STAR TOPOLOGY



C. BUS TOPOLOGY



D. RING TOPOLOGY



TOPOLOGIES (PART 3)

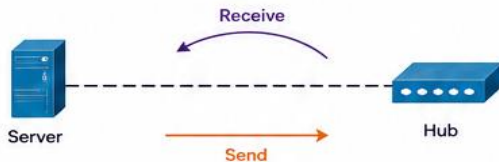
DUPLEX COMMUNICATION & ACCESS CONTROL METHODS

1. DUPLEX COMMUNICATION

In LANs/WLANs, data can flow in two directions between two devices. There are two modes of duplex communication.

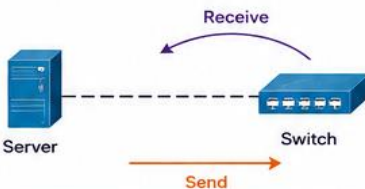
1.1 HALF-DUPLEX COMMUNICATION

- Nodes can send or receive, but not both at the same time.
- A device must wait for the medium to be free.
- Common in hub-based LANs.



1.2 FULL-DUPLEX COMMUNICATION

- Nodes can send and receive at the same time.
- More efficient and faster.
- Common in switch-based LANs.



KEY POINT

- Half-duplex = one direction at a time.
- Full-duplex = both directions at the same time.

2. ACCESS CONTROL METHODS

Access control decides how devices share the physical medium. Two basic methods for shared media:



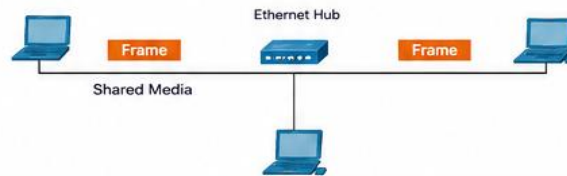
Contention-based access



Controlled access

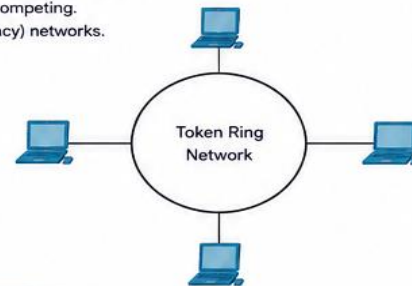
2.1 CONTENTION-BASED ACCESS

- Nodes operate in half-duplex (only one can transmit at a time).
- Devices compete for the medium.
- Process: More than one device wants to transmit → collision occurs → collision detection/avoidance is used.
- Examples:
 - Carrier Sense Multiple Access with Collision Detection (CSMA/CD) used in legacy bus Ethernet LANs
 - Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) used in Wireless LANs



2.2 CONTROLLED ACCESS

- Each node takes its turn to use the medium.
- Prevents collisions — no competing.
- Used in deterministic (legacy) networks.
- Examples:
 - Legacy Token Ring
 - Legacy ARCNET



3. CONTENTION-BASED ACCESS – CSMA/CD

Used in half-duplex Ethernet networks (e.g., old hubs). Only one device can send at a time.

Process:

- A device listens to the medium.
- If medium is idle → send frame.
- If collision occurs → both devices detect it.
- Both stop → wait random time → try again.



NOTE

Point-to-point over Ethernet:
Device must know if the incoming frame is for it. Nodes may be connected via intermediate devices (hubs, switches), but logical connection is still point-to-point.

1. PC1 Sends a Frame

PC1 has a frame to send to PC3. It listens to the medium. If idle → starts sending.



2. Hub Receives

Hub (repeater) receives the frame.



3. Hub Sends to PC3

Hub forwards the frame to all ports.



Legacy Ethernet (Wired)

4. CONTENTION-BASED ACCESS – CSMA/CA

Used in wireless LANs.

Similar to CSMA/CD but uses collision avoidance.

Process:

- Device listens to the medium.
- If medium is busy → waits for a random time.
- When medium is free → sends frame.
- Receiver sends acknowledgment (ACK).
- Sender knows frame was received.



NOTE

Ethernet LANs with switches use full-duplex mode, so CSMA/CD is not used (no collisions). Switch and host NIC operate in full-duplex.



★ KEY TAKEAWAY

- Half-duplex → one direction at a time | Full-duplex → both directions simultaneously
- Contention-based → compete for medium (CSMA/CD, CSMA/CA)
- Controlled access → take turns (Token Ring, ARCNET)

DATA LINK FRAME (PART 1)

1. THE FRAME – OVERVIEW



- The **Data Link Layer** prepares the encapsulated data (usually an IPv4 or IPv6 packet) for transport across the local media.



- It encapsulates the data with a **header** at the beginning and a **trailer** at the end to create a frame.



- The Data Link Protocol is responsible for **NIC-to-NIC communication** within the same network.



- Although many data link layer protocols exist, each frame type has three basic parts:

Header

Data

Trailer



- Unlike other layers, the Data Link Layer **appends information as a trailer** at the end of the frame.



- All data link layer protocols encapsulate the data within the **data field** of the frame.



- The structure of the frame and the fields in the header and trailer vary according to the protocol.



- There is no single frame structure that fits all media types and environments.

– The amount of control information in the frame varies based on:

- Media access control requirements
- Logical topology



- In environments with more errors or collisions (fragile environments), **more controls** are required to ensure delivery.

- Therefore, header and trailer fields are larger when more control information is needed.

2. FRAME FIELDS



- Framing breaks the stream into decodable groupings with **control information** in the header and trailer as values in different fields.

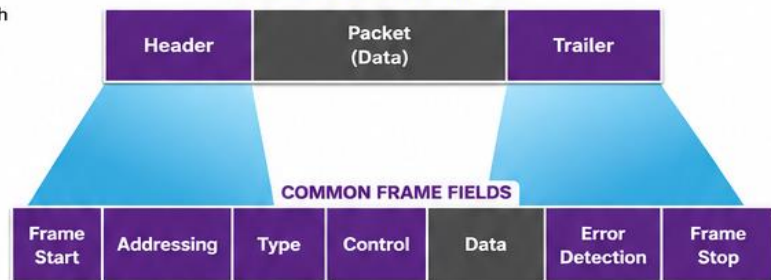


- This format gives physical signals a structure that is recognized by nodes and decoded into packets at the destination.



- The generic frame fields are shown on the right.
- Not all protocols include all these fields. The standards for a specific data link protocol define the actual frame format.

GENERAL FRAME STRUCTURE



DESCRIPTION OF COMMON FRAME FIELDS

 Frame Start and Stop Indicator flags	Used to identify the beginning and end limits of the frame.	 Data	Contains the frame payload (i.e., packet header, segment header, and the data).
 Addressing	Indicates the source and destination nodes on the media.	 Error Detection	Included after the data to form the trailer.
 Type	Identifies the Layer 3 protocol in the data field.	 Frame Stop	Marks the end of the frame.
 Control	Identifies specific control services such as Quality of Service (QoS). QoS gives forwarding priority to certain types of messages. For example, VoIP frames usually receive priority because they are sensitive to delay.		



ERROR DETECTION (TRAILER FUNCTION)

Data link protocols add a trailer to each frame. This trailer performs error detection. It determines whether the frame arrived without errors.

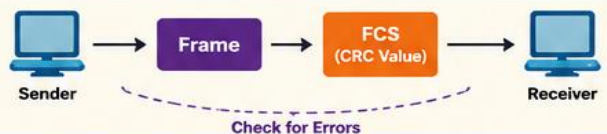
It generates a logical or mathematical summary (Checksum/CRC) of the frame bits.



CRC AND FCS (EXAMPLE IN ETHERNET)

The transmitting node creates a value called CRC (Cyclic Redundancy Check). This value is placed in the Frame Check Sequence (FCS) field in the trailer.

In Ethernet, the FCS helps the receiving node determine whether the frame experienced transmission errors.



KEY TAKEAWAY



- Frame = Header + Data + Trailer



Data link layer ensures NIC-to-NIC delivery on local media.



Frame fields vary by protocol and environment.



Trailer provides error detection (CRC/FCS).



More control info needed in fragile environments.

DATA LINK FRAME (PART 2)



1. LAYER 2 ADDRESSES

The data link layer uses addressing to transport a frame across a shared local media.



Device addresses at this layer are called **physical addresses (MAC addresses)**.



Layer 2 addressing is inside the **frame header**.



It specifies the **frame destination node** on the local network.

ADDRESSING RULES

- Unlike Layer 3 (logical) addresses, physical addresses (MAC) do not indicate where the device is located.
- When a frame moves to another network, the **MAC addresses change at each hop**, even if the same Layer 2 addresses are used in both networks.
- Layer 2 addresses are only used to connect devices **within the same shared media** on the same network.

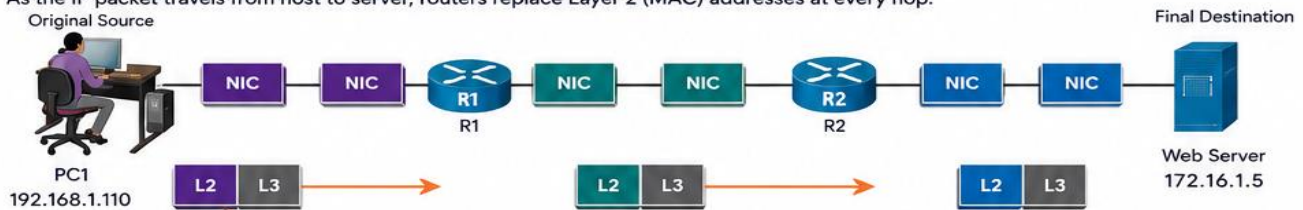


KEY POINT

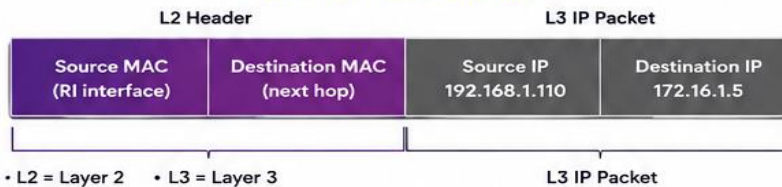
- Layer 2 addresses = used for local delivery only.
- They have no meaning beyond the local network.
- Layer 3 (IP) addresses = used for end-to-end delivery across networks.
- Router replaces Layer 2 addresses at every hop.

HOW LAYER 2 ADDRESSES WORK (EXAMPLE)

As the IP packet travels from host to server, routers replace Layer 2 (MAC) addresses at every hop.



A DATA LINK FRAME CONTAINS:



IN SUMMARY

- Each hop (router) creates a new L2 frame.
- New source & destination MAC addresses are used on each link.
- The IP addresses remain the same end-to-end.

2. LAN AND WAN FRAMES

Ethernet is used in wired LANs. Wireless uses WLAN (IEEE 802.11) protocols designed for multiaccess networks.

2.1 COMMON WAN PROTOCOLS (OVER DIFFERENT TOPOLOGIES)



Point-to-Point Protocol (PPP)
– Used for simple point-to-point links.



High-Level Data Link Control (HDLC)
– Used for reliable data transfer.



Frame Relay
– Uses virtual circuits over a shared network.



Asynchronous Transfer Mode (ATM)
– High-speed, connection-oriented technology.



X.25
– Older packet-switching protocol.

2.2 ABOUT LAYER 2 PROTOCOLS



In TCP/IP, all OSI Layer 2 protocols work with IP at OSI Layer 3.



Which Layer 2 protocol is used depends on the logical topology and the physical media.



Devices (switches, routers, access points, etc.) implement these Layer 2 protocols at the data link layer.



Layer 2 protocol choice depends on:

- ✓ Network size (number of hosts)
- ✓ Geographic scope (coverage area)
- ✓ Services required (bandwidth, latency, reliability)



KEY TAKEAWAY

LANs need high bandwidth for many hosts in a small area.
WANs cover large areas with fewer hosts—cost & bandwidth matter.

2.3 LAN AATA LINK PROTOCOLS (COMMONLY USED)

Protocol	Used In	Purpose/Feature
Ethernet	Wired LANs	Most common LAN technology.
IEEE 802.11 (Wi-Fi)	Wireless LANs (WLANs)	Supports multiaccess wireless access.
PPP	WANs (point-to-point)	Simple point-to-point connections.
HDLC	WANs (reliable links)	Bit-oriented, reliable data transfer.
Frame Relay	WANs	Efficient WAN with virtual circuits.



BIG PICTURE

- Layer 2 protocols are replaced in WANs by Ethernet.
- They provide access control and framing for the underlying topology.
- The right protocol ensures efficient, reliable communication based on network needs.